

Hexadirectional modulation of grid cell firing in rodents

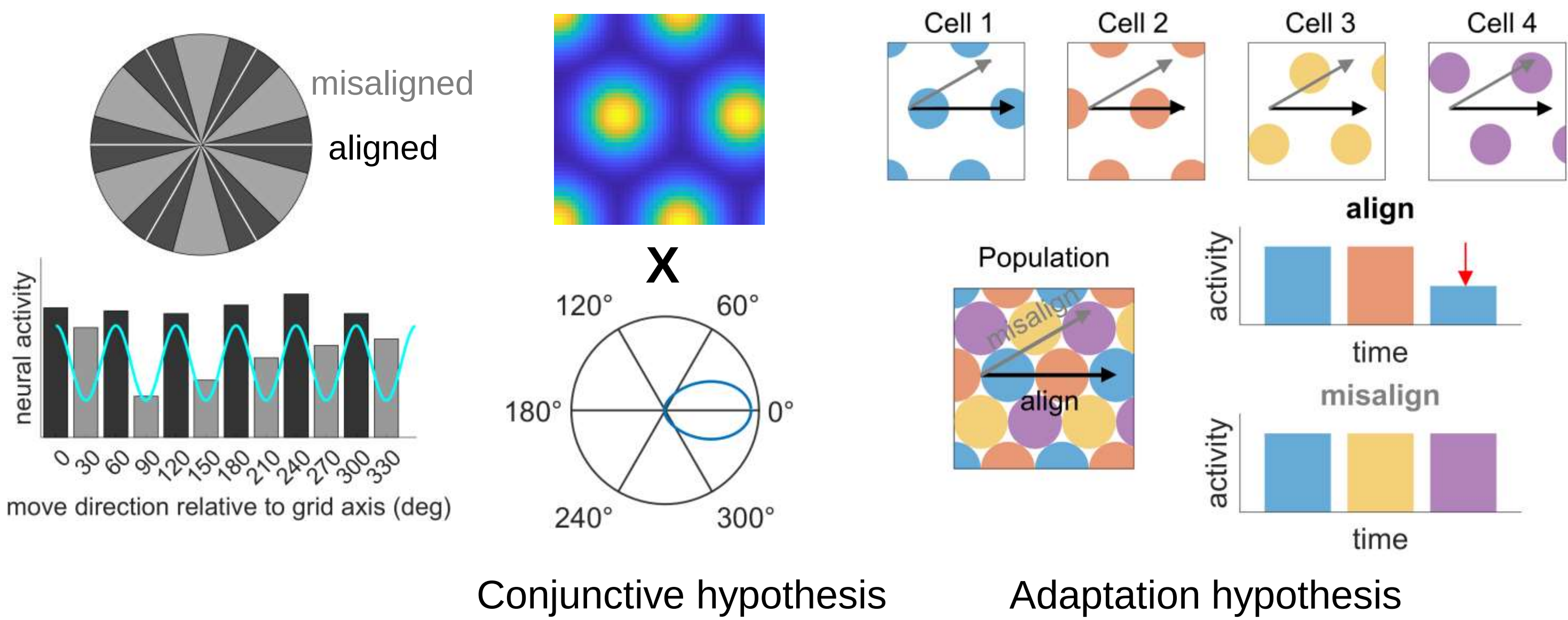
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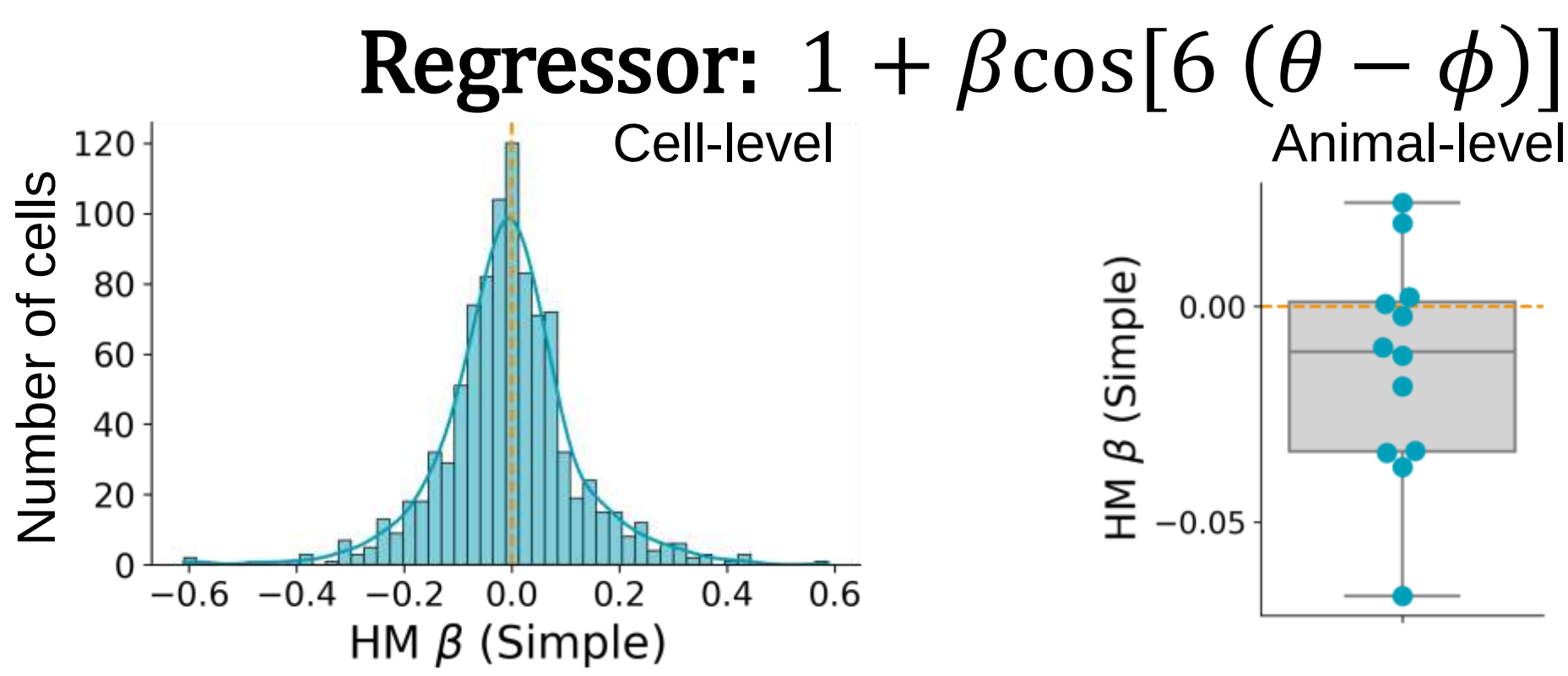
1. Introduction

- Macroscopic neural activities modulated by running direction (θ) with 6-fold symmetry were found in human fMRI and MEG [1,2].
- The **hexadirectional modulation (HM)** could arise from grid cell activity by two mechanisms [1,3].
- Grid x head direction (HD) conjunctive cell hypothesis:** the preferred HD is aligned to the grid axis.
- Adaptation hypothesis:** when the running direction is aligned to the grid axis, grid fields are visited at a regular interval, resulting in more firing rate adaptation compared to when the running direction is misaligned.
- We analysed a large dataset (~1,000 cells) of freely running rodents to test these hypotheses [4-7].

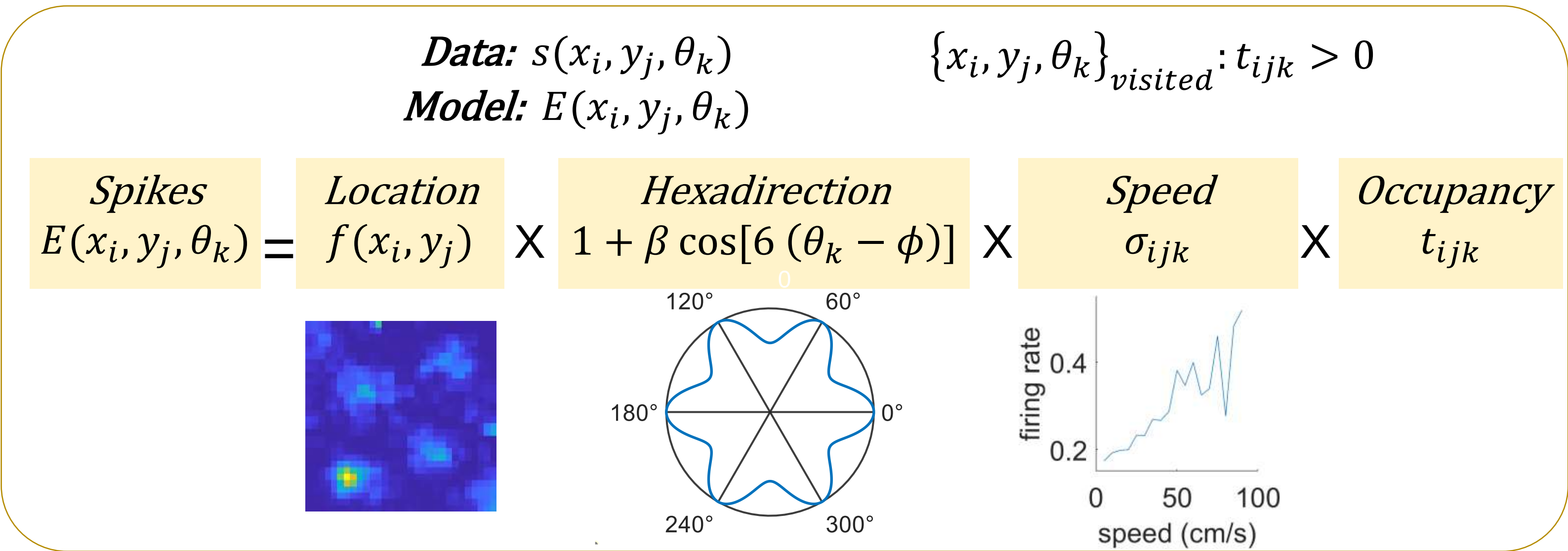


2. HM discovered when controlling for uneven sampling

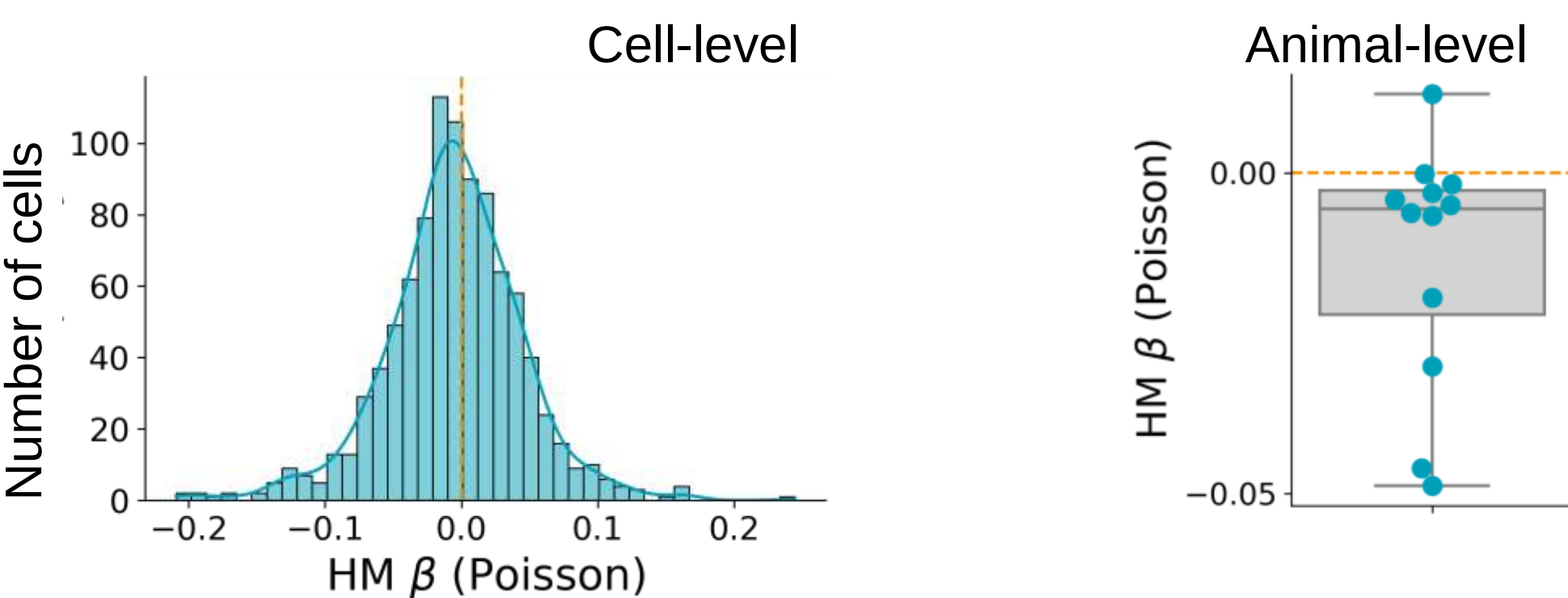
- We first applied a simple regression model to test whether individual cell's firing rate show HM. We found no significant modulation.



- Because uneven sampling of location, speed, and direction could hide weak HM in the data, we used a multiplicative Poisson firing model to account for this [7].

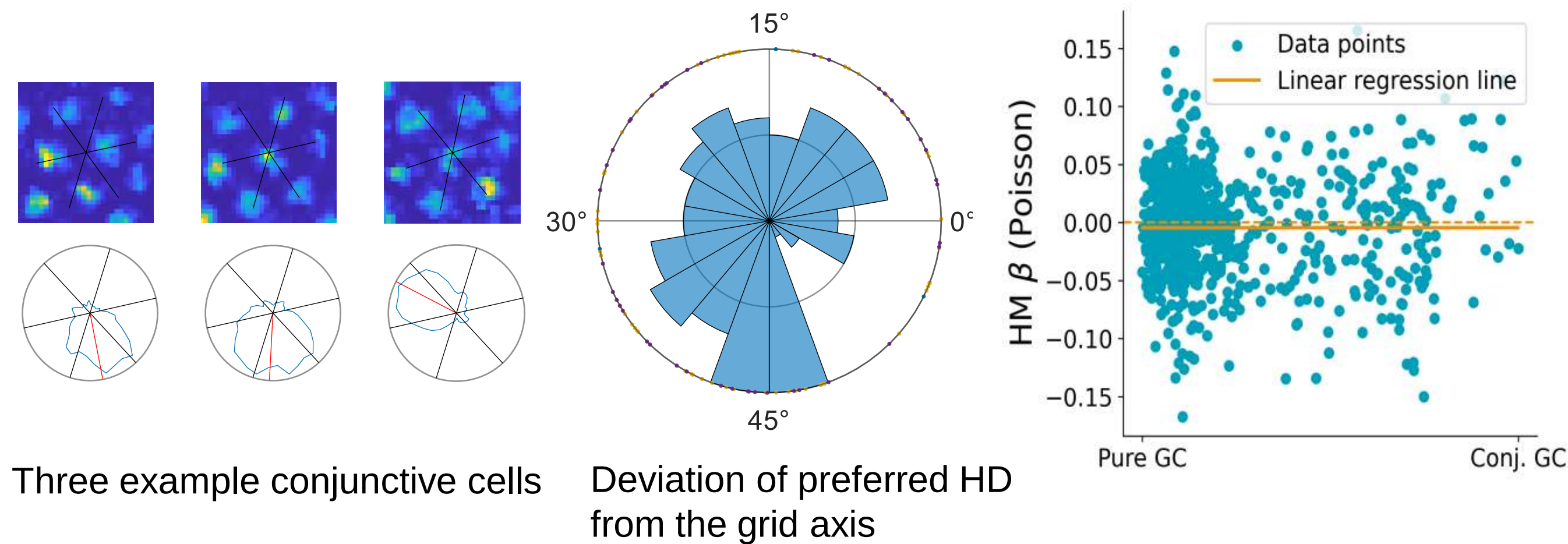


- We found a small, but significant negative beta, which means lower firing rate for aligned movement. This effect was specific to 6-fold symmetry.



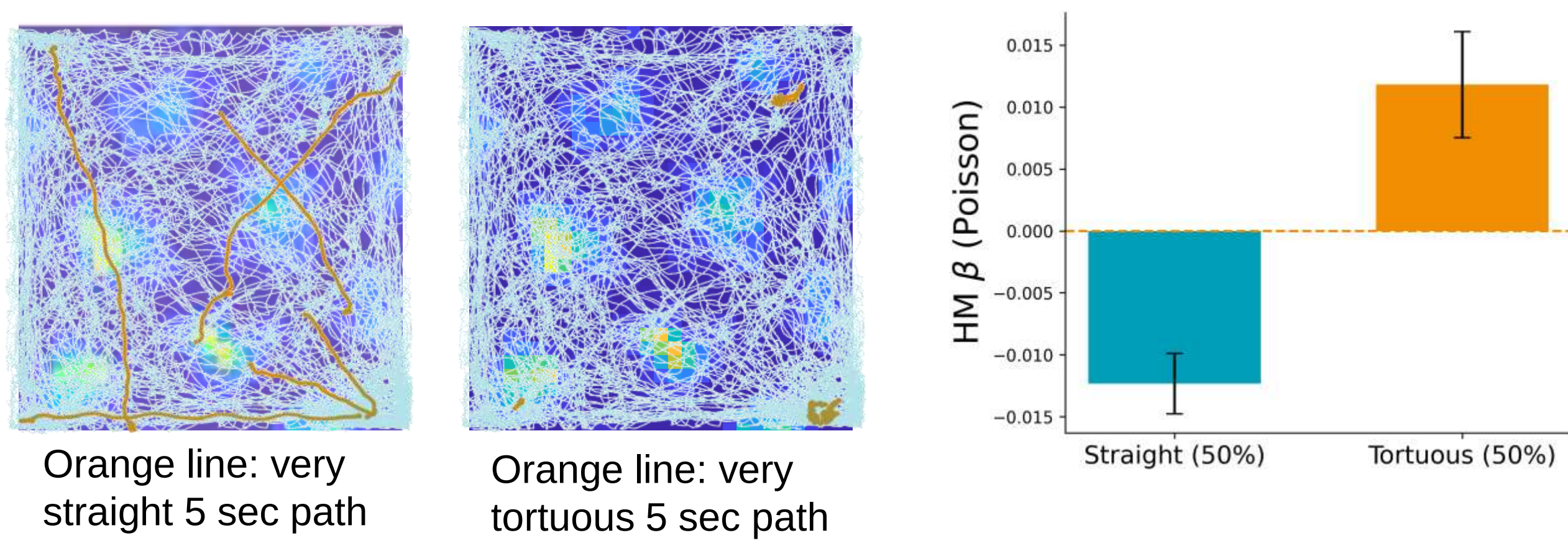
3. Conjunctive cells are not the source of HM

- HD tuning width is too broad for HM. ($80^\circ \pm 1^\circ$, $n=109$ cells)
- Preferred HD is not aligned to the grid axis.
- Conjunctive cells don't show stronger HM.

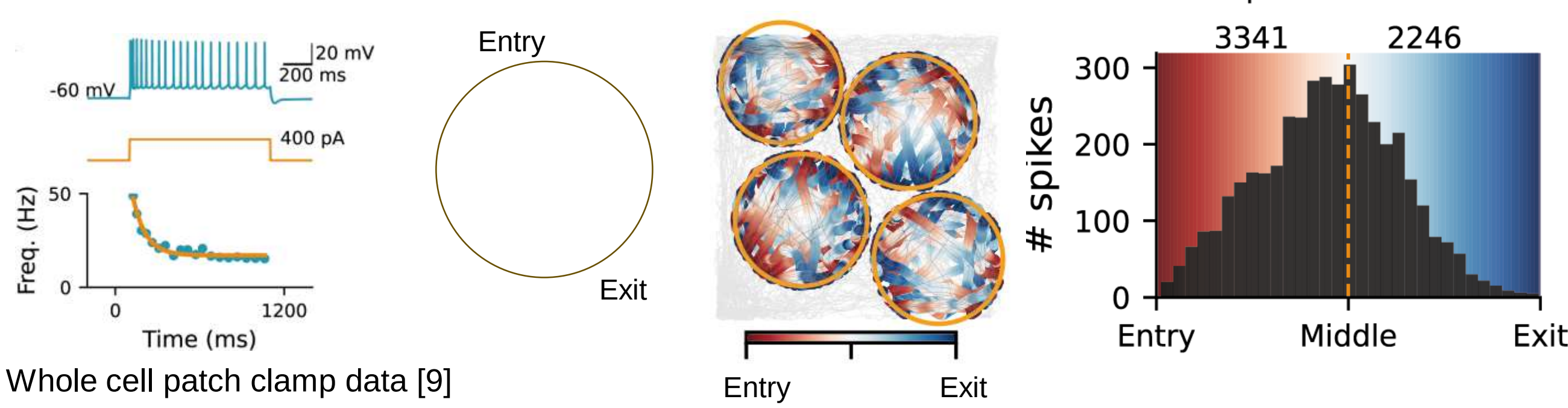


4. Adaptation is the likely source of HM

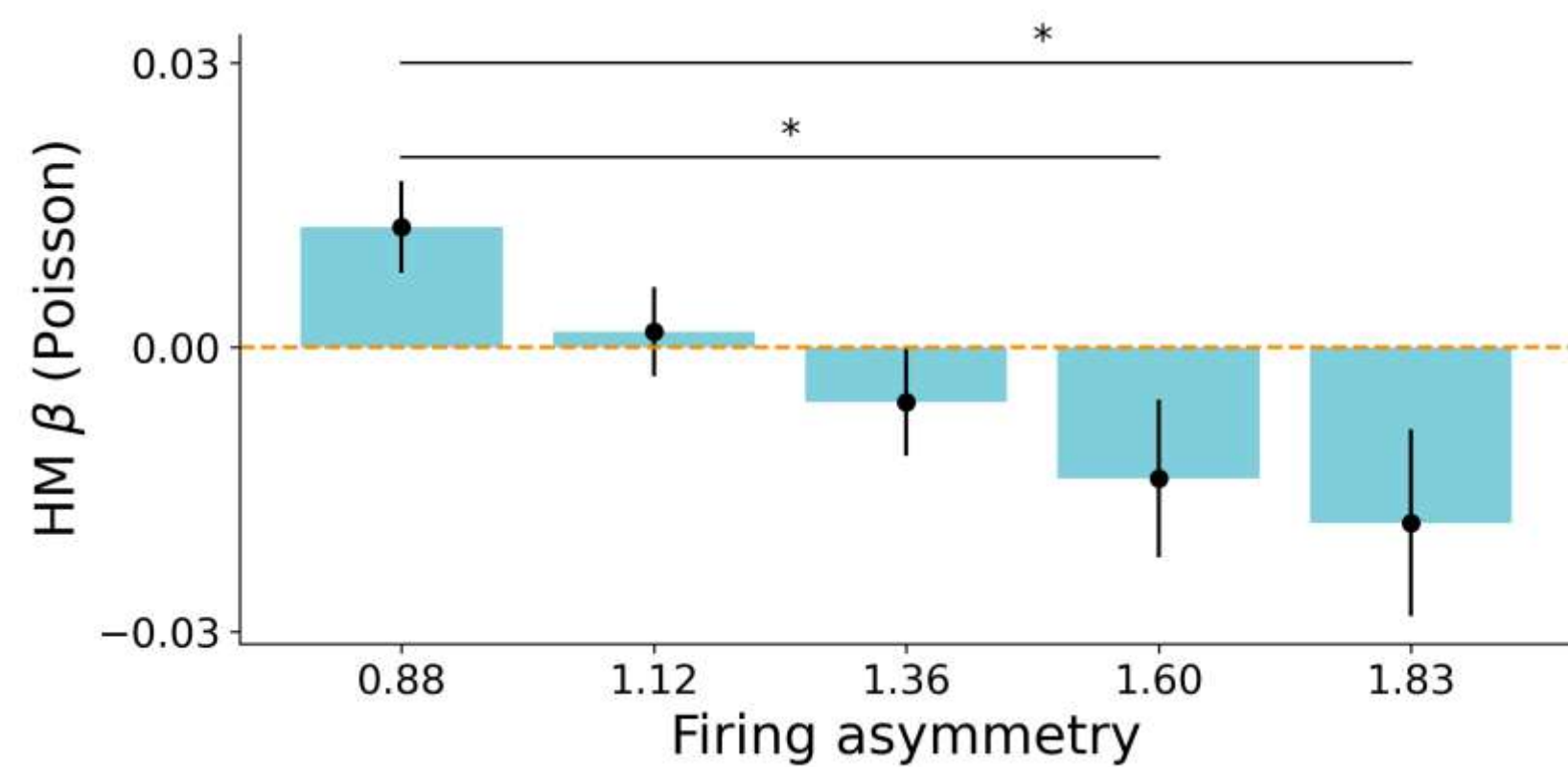
- Firing rate adaptation can induce HM when the animal runs fast and straight between neighbouring grid fields. If the trajectory is tortuous (e.g. circling within a small region), adaptation doesn't differ between the aligned and misaligned direction.
- Indeed, we found significant HM only in straighter paths.



- A key prediction of firing rate adaptation is asymmetric firing rates when the rat enters and exits a grid field (poster #33, Ji & Burgess)



- Grid cells which showed stronger asymmetric firing (more adaptation) showed stronger HM.



5. Discussion

- We provide the first evidence of HM at the single grid cell level and across animals during free foraging.
- We developed a Poisson regression model which accounts for uneven sampling, optimizing the HM detection.
- Firing rate adaptation seems to drive the HM, supported by model simulations and empirical data.
- We will further investigate the adaptation model (e.g. strength, time constant, grid scale) to optimise the experiments for HM detection.

References

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